Lab 1: Postural Assessments (RULA/REBA vs. Pose Estimation)

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Date: 17th February

Introduction

RULA and REBA are quick observational tools that flag awkward postures and the likelihood of musculoskeletal strain. In this lab, five participants completed two short tasks: a packaging task (Task 1, scored with RULA) and a cargo-handling task (Task 2, scored with REBA). For each video, I selected a representative "highest-risk" moment and scored posture using the appropriate worksheet. In parallel, a camera-based pose-estimation script extracted joint angles over time (trunk, neck, shoulder, elbow, knee, and optional wrist deviation), so I could quantify both typical posture (mean and variability) and exposure duration (percent time above risk thresholds). The goal was to see how well the quick single-frame scores line up with the angle-based measurements, and what extra insight the full time series provides.

Methods

I set up a simple workstation to mimic the two manual-handling tasks using a lightweight cardboard box and a table-height surface. For Task 1 (Packaging), each participant repeated a short packing cycle (reach, place, pack, and reposition) while staying in frame. For Task 2 (Cargo), they lifted and transferred the box between two marked locations to create bending and reaching demands. A second person recorded each task from a steady side/oblique view with the full body visible, then I trimmed the clips to the active task segment.

For scoring, I paused each clip at the posture that looked most demanding for the target assessment: upper-body strain for RULA (Task 1) and whole-body strain for REBA (Task 2), and scored using the standard worksheets. I then ran the same trimmed videos through the provided camera-based pose-estimation script to extract per-frame joint angles (trunk, neck, shoulder, elbow, wrist, knee). From the angle output, I calculated mean, standard deviation, and maximum values, along with temporal exposure metrics (% time trunk flexion >20° and >60°, and % time

shoulder flexion >45°). Finally, I compared observational scores to the angle-based metrics across participants using Pearson's r.

Results

Video frames used for scoring

Participant	Task	Video file	Timestamp	Frame
P1	Task 1 – Packaging (RULA)	SR_Task2_1.MOV.mp4	00:00:02	65
P1	Task 2 – Cargo (REBA)	SR_Task2_2.MOV	00:00:06	399
P2	Task 1 – Packaging (RULA)	MLTASK1.2.MOV	00:00:03	109
P2	Task 2 – Cargo (REBA)	MLTASK2.2.MOV.mp4	00:00:01	48
P3	Task 1 – Packaging (RULA)	MJ_Task1.2.MOV.mp4	00:00:02	76
P3	Task 2 – Cargo (REBA)	MJ_Task_2.MOV	00:00:01	58
P4	Task 1 – Packaging (RULA)	SN_TASK1.MOV	00:00:01	58
P4	Task 2 – Cargo (REBA)	SN_TASK2.MOV	00:00:03	98
P5	Task 1 – Packaging (RULA)	DM_1_1.MOV	00:00:02	83
P5	Task 2 – Cargo (REBA)	ER2_2.MOV.mp4	00:00:03	98

Table 1. Timestamp and frame selected from each video for the single-frame RULA/REBA scoring.

Observational scores (RULA/REBA)

Participant	Task 1 RULA score	RULA action level	Task 2 REBA score	REBA risk level
P1	4	Action Level 2 (investigate further)	7	Medium risk
P2	3	Action Level 2 (investigate further)	6	Medium risk
P3	3	Action Level 2 (investigate further)	5	Medium risk
P4	3	Action Level 2 (investigate further)	7	Medium risk
P5	4	Action Level 2 (investigate further)	5	Medium risk

Table 2. RULA (Task 1) and REBA (Task 2) scores by participant, with standard action/risk categories.

Pose estimation summary: Task 1 – Packaging (RULA)

Joint/Segment Angle	Mean (SD) –P1	Max–P1	Mean (SD) –P2	Max–P2	Mean (SD)–P3	Max–P3	Mean (SD) –P4	Max–P4	Mean (SD) –P5	Max–P5	Mean (SD) of Mean–All	Mean (SD) of Max–All

Trunk Flexion Angle (degrees)	11.49 (4.42)	21.59	6.52 (3.68)	16.66	12.05 (4.35)	18.25	12.49 (4.02)	21.61	20.88 (4.97)	29.43	12.69 (5.17)	21.51 (4.92)
Neck Flexion Angle (degrees)	60.49 (3.55)	66.70	55.02 (2.97)	59.78	64.37 (3.25)	71.98	58.64 (3.31)	64.16	65.72 (4.10)	71.91	60.85 (4.33)	66.91 (5.22)
Shoulder Flexion Angle (degrees)	27.15 (12.79)	58.56	19.90 (10.60)	42.45	22.67 (6.03)	31.53	22.58 (7.94)	47.17	30.28 (7.72)	45.42	24.52 (4.14)	45.03 (9.70)
Elbow Flexion Angle (degrees)	53.04 (15.32)	83.46	41.45 (18.23)	92.33	44.35 (8.14)	64.66	27.22 (10.84)	49.82	49.14 (12.43)	74.96	43.04 (9.90)	73.05 (16.54)
Knee Flexion Angle (degrees)	23.78 (3.45)	34.47	28.65 (2.03)	34.15	32.84 (4.63)	49.23	28.22 (2.48)	36.24	28.76 (2.52)	34.79	28.45 (3.21)	37.78 (6.45)

Table 3. Pose-estimation joint angles for Task 1 (Packaging). Values are mean (SD) across frames and the maximum observed angle in the clip.

Temporal exposure: Task 1 – Packaging

Temporal/Postural Metric	**P1**	**P2**	**P3**	**P4**	**P5**	**Mean (SD)**
Percentage of time trunk flexion > 20°	5.06%	0.00%	0.00%	4.95%	57.14%	13.43% (24.56%)

Percentage of time trunk flexion > 60°	0.00%	0.00%	0.00%	0.00%	0.00%	0.00% (0.00%)
Percentage of time shoulder flexion > 45°	15.19%	0.00%	0.00%	5.86%	1.19%	4.45% (6.47%)

Table 4. Temporal exposure metrics for Task 1 (Packaging): percentage of frames above common ergonomic threshold angles.

Pose estimation summary: Task 2 – Cargo handling (REBA)

Joint/Segment Angle	**Mean (SD) –P1**	**Max–P1**	**Mean (SD) –P2**	**Max–P2**	**Mean (SD) –P3**	**Max–P3**	**Mean (SD) –P4**	**Max–P4**	**Mean (SD)–P5**	**Max–P5**	**Mean (SD) of Mean–All**	**Mean (SD) of Max–All**
Trunk Flexion Angle (degrees)	13.64 (6.97)	34.11	16.03 (9.51)	37.39	10.43 (7.46)	27.08	18.22 (10.81)	35.23	15.58 (5.50)	24.31	14.78 (2.93)	31.62 (5.63)
Neck Flexion Angle (degrees)	60.19 (3.97)	67.33	53.14 (3.69)	63.41	55.99 (4.49)	65.97	58.73 (4.08)	66.20	61.32 (4.88)	67.84	57.87 (3.31)	66.15 (1.72)
Shoulder Flexion Angle (degrees)	26.71 (16.42)	71.01	21.95 (11.28)	38.17	19.92 (13.39)	49.20	29.03 (17.17)	51.80	19.86 (9.28)	42.73	23.49 (4.16)	50.58 (12.61)
Elbow Flexion Angle (degrees)	46.18 (11.57)	80.60	43.40 (21.11)	89.70	33.50 (19.50)	66.50	36.95 (15.84)	66.09	30.73 (7.87)	51.44	38.15 (6.52)	70.87 (14.74)
Knee Flexion Angle (degrees)	23.08 (4.96)	39.37	23.28 (3.73)	31.60	19.12 (4.58)	28.42	21.71 (5.68)	32.69	18.53 (6.18)	30.70	21.14 (2.21)	32.56 (4.12)

Table 5. Pose-estimation joint angles for Task 2 (Cargo). Values are mean (SD) across frames and the maximum observed angle in the clip.

Temporal exposure: Task 2 – Cargo handling

Temporal/Postural Metric	**P1**	**P2**	**P3**	**P4**	**P5**	**Mean (SD)**
Percentage of time trunk flexion > 20°	13.35%	33.06%	15.24%	42.86%	30.16%	26.93% (12.48%)
Percentage of time trunk flexion > 60°	0.00%	0.00%	0.00%	0.00%	0.00%	0.00% (0.00%)
Percentage of time shoulder flexion > 45°	13.35%	0.00%	5.49%	35.17%	0.00%	10.80% (14.68%)

Table 6. Temporal exposure metrics for Task 2 (Cargo): percentage of frames above threshold angles.

Key observations from the pose-estimation data

For Task 1 (Packaging), neck flexion stayed high across participants (mean range 55.0° to 65.7°), which matches the "head-down" posture seen while looking into the box or at the work surface.

Trunk flexion varied a lot in Task 1: Participant P5 showed the largest average trunk flexion (20.9° mean) and also spent the most time above 20° trunk flexion (57.1% of the clip), suggesting sustained forward bending rather than a quick peak.

Wrist deviation was consistently large in both tasks (Task 1 mean ≈ 53.6°, Task 2 mean ≈ 54.0°), which likely reflects how the hands were oriented during grasping and placement.

For Task 2 (Cargo), shoulder flexion peaks were the biggest standout (maximum shoulder flexion up to 71.0°), especially during reaching and repositioning.

Correlation: RULA/REBA vs. pose estimation (Pearson r)

To meet the "at least 5 data points" requirement, I correlated each observational score (one per participant) with multiple pose-based metrics (means and temporal exposures).

Task 1 – Packaging (RULA)

Variable (Pose/Temporal)	Pearson r with RULA
Mean Trunk Flexion (deg)	0.617
Mean Neck Flexion (deg)	0.475
Mean Shoulder Flexion (deg)	0.925
Mean Elbow Flexion (deg)	0.743
Mean Knee Flexion (deg)	-0.620
% Time Trunk > 20°	0.657
% Time Shoulder > 45°	0.528

Table 7. Pearson correlation between Task 1 pose/temporal variables and RULA scores (n=5 participants).

Task 2 – Cargo (REBA)

Variable (Pose/Temporal)	Pearson r with REBA
Mean Trunk Flexion (deg)	0.500
Max Trunk Flexion (deg)	0.798
Mean Shoulder Flexion (deg)	0.958
Max Shoulder Flexion (deg)	0.612
% Time Trunk > 20°	0.217
% Time Shoulder > 45°	0.733

Table 8. Pearson correlation between Task 2 pose/temporal variables and REBA scores (n=5 participants).

Discussion

1. Comparison of RULA/REBA vs. pose-estimation angles

Overall, the single-frame RULA/REBA scores lined up reasonably well with what the pose-estimation data showed, but they are not measuring the exact same thing. RULA and REBA

were scored on one selected moment (the posture that looked worst), while pose estimation summarizes the whole clip (typical posture, variability, and peaks). For example, Task 1 had consistently high neck flexion across all participants, which helps explain why most RULA scores were in the 3–4 range. At the same time, participants who had higher average trunk or shoulder flexion and more time above thresholds tended to have slightly higher RULA/REBA scores. Where I noticed small mismatches, the most likely reason was frame selection: a brief extreme posture can increase a RULA/REBA score even if the average posture over the clip is moderate.

2. Temporal analysis benefits

The time-series output adds an exposure and variability story that the single-frame method cannot capture. In Task 1, time above 20° trunk flexion ranged from 0.0% to 57.1% across participants, so two people could score similarly on RULA but have very different total exposure. The same idea shows up in shoulder elevation: some participants had essentially no time above 45°, while others spent a noticeable portion of the clip there. This matters because the risk is often driven by how long the body stays in an awkward posture, not just how awkward it gets at its worst.

3. Advantages and limitations of each method

RULA/REBA are fast, cheap, and easy to apply in the field, and they translate posture into an action-oriented score that is easy to communicate. Their main downside is subjectivity: the score depends on the observer, the chosen frame, and how accurately loads/coupling are judged. Pose estimation is more objective and gives full temporal coverage (means, peaks, and percent-time exposure), but its accuracy depends on camera angle, occlusion, clothing, and how well the model tracks joints. It also does not directly capture important ergonomic factors like force, load asymmetry, grip quality, or contact stress unless those are added separately.

4. Integration of methods

The cleanest workflow would use both tools together. I would start with pose estimation to screen the full clip and automatically flag candidate “worst” frames (for example, the highest trunk or shoulder flexion, or the moment with the largest combined risk). Then I would perform RULA/REBA on those flagged frames to capture factors the camera cannot infer (load, coupling, and task context). If the RULA/REBA score disagreed with the pose-based picture, I would re-check whether the selected frame was representative or whether the pose model lost track of key joints. In other words, pose estimation is great for finding and quantifying the problem over time, and RULA/REBA is useful for translating that into a standard ergonomic action level.

5. Ergonomic intervention recommendations

Two changes would likely reduce risk for both tasks. First, raise and/or bring the work closer to neutral posture: adjusting table height, using a tilt bin, or repositioning the box so the worker

does not need to look down and reach forward as much would target the consistently high neck flexion observed in Task 1. Second, reduce elevated reaching during cargo handling by staging items at waist-to-chest height (use a height-adjustable cart, turntable, or intermediate shelf) so the shoulder does not repeatedly move into high flexion. Based on the pose results, several participants reached shoulder flexion maxima near or above ~80–90°, and the temporal metrics show non-trivial time above 45° for some trials. A smaller but practical addition would be wrist-neutral handling aids (better-shaped handles, different grasp strategy, or gloves that improve coupling), since wrist deviation stayed high in both tasks.